

CTD

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.2872 ±0.00605	0.5117 ±0.0171	0.3842 ±0.018	0.3219 ±0.00585
Cramér $\ell_2^2 \downarrow$	0.02843 ±0.00163	0.03076 ±0.00204	0.04107 ±0.00156	0.0295 ±0.000618
CF-$\ell_2^2/\omega^2 \downarrow$	0.02227 ±0.000497	0.02499 ±0.00139	0.03445 ±0.00183	0.02355 ±0.000736

QTD

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.2012 ±0.0202	0.4639 ±0.025	0.4028 ±0.0186	0.2512 ±0.0188
Cramér $\ell_2^2 \downarrow$	0.01011 ±0.0022	0.02363 ±0.0031	0.04633 ±0.0042	0.01378 ±0.00366
CF-$\ell_2^2/\omega^2 \downarrow$	0.009752 ±0.00218	0.0223 ±0.00305	0.04584 ±0.00423	0.01323 ±0.0033

ϕ TD Dirac

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.1517 ±0.022	0.439 ±0.0777	0.2514 ±0.107	0.1665 ±0.0163
Cramér $\ell_2^2 \downarrow$	0.00665 ±0.00295	0.01542 ±0.00499	0.01357 ±0.0128	0.00466 ±0.00113
CF-$\ell_2^2/\omega^2 \downarrow$	0.00579 ±0.00272	0.01293 ±0.00448	0.0124 ±0.0125	0.003824 ±0.00102

ϕ TD Cat

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.2924 ±0.00677	0.4465 ±0.0133	0.3722 ±0.0424	0.3182 ±0.00635
Cramér $\ell_2^2 \downarrow$	0.02889 ±0.00144	0.02447 ±0.00198	0.03646 ±0.00183	0.02855 ±0.00111
CF-$\ell_2^2/\omega^2 \downarrow$	0.0226 ±0.000894	0.02073 ±0.00178	0.03143 ±0.00236	0.02281 ±0.00121

ϕ TD Quant

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.1509 ±0.032	0.3711 ±0.0201	0.1892 ±0.0179	0.1575 ±0.015
Cramér $\ell_2^2 \downarrow$	0.006027 ±0.00313	0.009432 ±0.00168	0.007823 ±0.00134	0.003725 ±0.00117
CF-$\ell_2^2/\omega^2 \downarrow$	0.005385 ±0.00297	0.008582 ±0.00174	0.007326 ±0.0013	0.003098 ±0.00112